

Module 1: Introduction : Evolution of OR, Definition, scope of Operations Research, characteristics, Advantages and limitations. Quantitative approach to decision making models (Theory Only).

I. MEANING AND DEFINITION

1. Operations research is the art of winning wars without actually fighting them.
2. Operations research is the art of finding bad answers to problems which otherwise have worse answers.
3. Operations research is concerned with scientifically deciding how to best design and operate man machine systems that usually require the allocation of scarce resources – **J K Sharma**.
4. Operations research is the systematic application of quantitative methods, techniques and tools to the analysis of problems involving the operation of systems. – **Daellenbach and George**.
5. Operations research is a scientific approach to problem-solving for executive management – **H M Wagner**.

II. SCOPE OF OPERATIONS RESEARCH

1. In Agriculture

- Optimum allocation of land to various crops in accordance with the climatic conditions.
- Optimum distribution of water from various resources like canal for irrigation purposes

2. In Finance

- To maximize the per capita income with minimum resources
- To find out the profit plan for the company.
- To determine the best replacement policies etc.

3. In Industry- OR Is useful to the Industry Director in deciding optimum allocation of various limited resources such as men machine money time etc. to arrive at the optimum decision.

4. In Marketing

- Where to distribute the products for sale so that the total cost of transportation etc. is minimum.
- The minimum per unit sale price. The size of the stock to meet future demand.
- How to select the best advertising media with respect to time & cost
- How when & what to purchase at the minimum possible cost
- In personnel Management To appoint the most suitable persons on minimum salary
- To determine the best age of retirement for the employees.
- To find out the number of persons to be appointed on fulltime basis when the workload is seasonal.

6. In production Management

- To find out the number & size of the items to be produced.
- In scheduling & sequencing the production runs by proper allocations of machines.
- In calculating the optimum product Mix.
- To select, locate & design the sites for the production plants

7. In LI C

- What should be the premium rates for various modes of policies.
- How best the profit could be distributed in the cases of with Profit policies

8. Health care management and hospital administration
9. Government departments or public sectors
10. Energy and environment
11. Telecommunication network
12. Military defences
13. Supply chain and logistics management
14. National plans

III. CHARACTERISTICS AND FEATURES OF OR

1. Interdisciplinary,
2. Scientific,
3. Holistic, and
4. Objective oriented approaches to decision making

OR utilizes a planned approach following a scientific method and an interdisciplinary team, in order to represent complex functional relationship as mathematical models, for the purpose of providing a quantitative basis for decision-making and uncovering new problems for quantitative analysis. This definition implies additional features of OR approach. The broad features of OR approach in solving any decision problem are summarized as follows:

1. **Interdisciplinary approach:** For solving any managerial decision problem often an interdisciplinary teamwork is essential. This is because while attempting to solve a complex management problem, one person may not have the complete knowledge of all its aspects such as economic, social, political, psychological, engineering, etc. Hence, a team of individuals specializing in various functional areas of management should be organized so that each aspect of the problem can be analysed to arrive at a solution acceptable to all sections of the organization.
2. **Scientific approach:** Operations research is the application of scientific methods, techniques and tools to problems involving the operations of systems so as to provide those in control of operations with optimum solutions to the problems. The scientific method consists of observing and defining the problem; formulating and testing the hypothesis; and analysing the results of the test. The data so obtained is then used to decide whether the hypothesis should be accepted or not. If the hypothesis is accepted, the results should be implemented, otherwise not.
3. **Holistic approach:** While arriving at a decision, an operations research team examines the relative importance of all conflicting and multiple objectives. It also examines the validity of claims of various departments of the organization from the perspective of its implications to the whole organization.
4. **Objective-oriented approach:** An operations research approach seeks to obtain an optimal solution to the problem under analysis. For this, a measure of desirability (or effectiveness) is defined, based on the objective(s) of the organization. A measure of desirability so defined is then used to compare alternative courses of action with respect to their possible outcomes.

IV. OR APPROACH TO PROBLEM SOLVING / PHASES OF OR

Operations research approach to decision making is based on three phases

1. Judgement,
2. Research, and
3. Action

1. Judgement phase: This phase includes

- Identification of the real-life problem,
- Selection of an appropriate objective and the values of various variables related to this objective,
- Application of the appropriate scale of measurement, i.e. deciding the measures of effectiveness (desirability), and
- Formulation of an appropriate model of the problem and the abstraction of the essential information, so that a solution to the decision-maker's goals can be obtained.

2. Research phase: This phase is the largest and longest amongst all the phases. However, even though the remaining two are not as long, they are also equally important as they provide the basis for a scientific method. This phase utilizes:

- Observations and data collection for a better understanding of the problem,
- Formulation of hypothesis and model,
- Observation and experimentation to test the hypothesis on the basis of additional data,
- Analysis of the available information and verification of the hypothesis using pre-established measures of desirability,
- Prediction of various results from the hypothesis, and
- Generalization of the result and consideration of alternative methods.

3. Action phase: This phase consists of making recommendations for implementing the decision. This decision is implemented by an individual who is in a position to implement results. This individual must be aware of the environment in which the problem occurred, be aware of the objective, of assumptions behind the problem and the required omissions of the model.

V. ADVANTAGES OF OPERATIONS RESEARCH STUDY

In addition to learning about various OR models that are useful to arrive at an optimal decision, the knowledge of operations research techniques can also help the decision-maker to attain several other advantages as listed below.

- 1. Structured approach to Problems:** A substantial amount of time and effort can be saved in developing and solving OR models if a logical and consistent approach is followed. This implies that the decision maker has to be careful while defining variables, availability of resources, and functional relationships among variables in the objective function and constraints. This will also reduce the chance of conceptual and computational errors. Any such error can also be detected easily and corrected at an early stage.
- 2. Critical approach to problem solving:** The decision-maker will come to understand various components of the problem and accordingly select a mathematical model for solving the given problem. He will become aware of the explicit and implicit assumptions and limitations of such

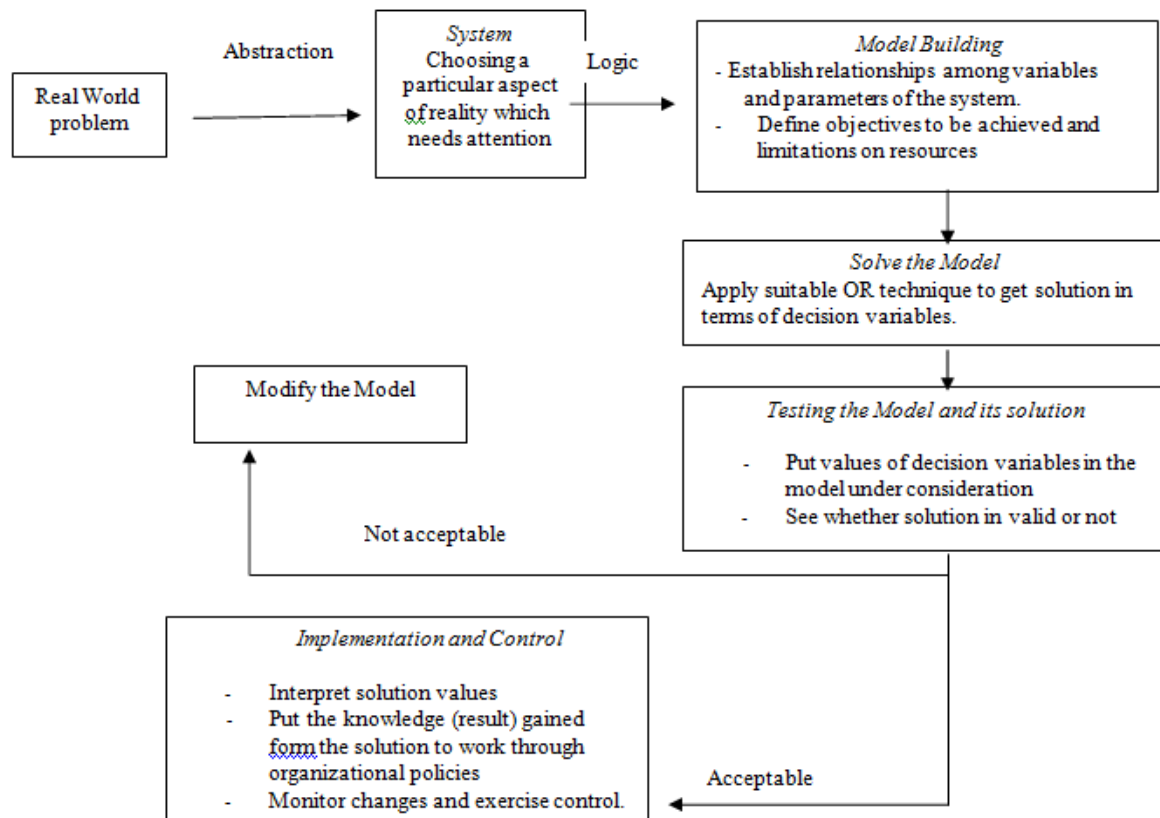
models. Problem solutions are examined critically and the effect of any change and error in the problem data can be studied through sensitivity analysis techniques.

3. It compels the decision-maker to be quite explicit about his objective assumptions and his perspective to constraints.
4. It makes the decision-maker very carefully consider exactly what variables influence decisions.
5. Quickly point out gaps in the data required to support solutions to the problem.
6. Its models can be solved by a computer, thus the management can get enough time for decisions that require a quantitative approach.
7. Better systems
8. Better control
9. Better coordination

VI. DISADVANTAGES OF OR

1. The solution to a problem is often derived by simplifying assumptions. Consequently, such solutions are not optimal.
2. Models may not represent the realistic situations in which decisions are made.
3. Often the decision-maker is not fully aware of the limitations of the models.
4. Many real world problems just cannot have an OR solution.
5. Dependence on electronic computer.
6. Non quantifiable factors.
7. Distance between manager and operations researcher.
8. Money and time cost.

VII. METHODOLOGY OF OPERATIONS RESEARCH



Every OR specialist may have his/her own way of solving problems. However, the effective use of OR techniques requires to follow a sequence of steps.

Step 1: Defining the problem: Problem definition involves the process of identifying, understanding, and describing the problem being faced by an organization. It helps to define the objective(s) to be achieved, and the alternative courses of action. The procedure begins with gathering information (data) of the organizational structure, communication and control system, objectives and expectations. Such information will help in assessing the difficulty likely to be faced in terms of costs, time, resources, probability of success of the study, etc.

Step 2: Formulating a mathematical model: After the problem is clearly defined and understood, the next step is to collect required data and then *formulate a mathematical model*. Model formulation requires defining relationships among decision variables.

Step 3: Solving the mathematical model: Once a mathematical model of the problem has been formulated, the next step is to obtain numerical values of decision variables. Obtaining these values depends on the specific form or type of mathematical model. In general, the following two categories of methods are used for solving an OR model.

- *Analytical Methods*
- *Heuristic Method*

Step 4: Validating the solution: After solving the mathematical model, it is important to review the solution to see whether value of variables makes sense and that the resulting decisions can be implemented. Some of the reasons for validating the solution are:

- Mathematical model may not have considered all constraints present in study because few constraints may either be omitted or incorrect.
- Certain aspects of the problem may have been overlooked, omitted or simplified.
- The data may have been incorrectly estimated or recorded.

Step 5: Implementing the solution: Before implementing the solution, the decision-maker should ensure to select alternatives that are capable of being implemented. Further, it is also important to ensure that any solution implemented is continuously reviewed and updated in the light of a changing environment. In any case, the decision-maker who is in the best position to implement solution (or result), must be aware of the objective, assumption, omissions and limitations of the model.

Step 6: Modifying the model: For a mathematical model to be useful, the degree to which it actually represents the system or problem being modelled must be established. If during validation the solution cannot be implemented, then identify the constraint(s) that were omitted during the mathematical model formulation or find out whether some of the original constraints were incorrect. In all such cases, return to the problem formulation step and carefully make the appropriate modifications to represent the given problem more accurately. A model must be applicable for a reasonable time period and should be updated from time to time, taking into consideration the past, present and future aspects of the problem.

Step 7: Establishing controls over the solution: The changes within the environment (society or business) influence the continuing validity of models and their solutions. Thus, a control procedure needs to be developed for detecting significant changes in decision variables of the problem, without having to build a model every time a significant change occurs.

VIII. MODELS OF OR AND MODELLING IN OPERATIONS RESEARCH

The key to model-building lies in abstracting only the relevant variables that affect the criteria of the measures-of-performance of the given system and in expressing the relationship in a suitable form. However, a model should be as simple as possible so as to give the desired result. On the other hand, oversimplifying the problem can also lead to a poor decision. Model enrichment is done by changing value of variables, and relaxing assumptions. The essential three qualities of any model are:

- **Validity** of the model – model should represent the critical aspects of the system/problem under study,
- **Usability** of the model – a model can be used for the specific purposes, and
- **Value** of the model to the user.

Besides these three qualities, other consideration of interest is,

- Cost of the model and its sophistication,
- Time involved in formulating the model, etc.

A. CLASSIFICATION BASED ON STRUCTURE

1. Physical models: These models are used to represent the physical appearance of the real object under study, either reduced in size or scaled up. Physical models are useful only in design problems because they are easy to observe, build and describe. For example, in the aircraft industry, scale models of a proposed new aircraft are built and tested in wind tunnels to record the stresses experienced by the air frame. Physical models cannot be manipulated and are not very useful for prediction. Problems such as portfolio selection, media selection, production scheduling, etc., cannot be analysed with the help of these models.

Physical models are classified into two categories.

(i) Iconic Models: An iconic model is a scaled (small or big in size) version of the system. Such models retain some of the physical characteristics of the system they represent. Examples of iconic model are, blueprints of a home, maps, globes, photographs, drawings, air planes, trains, etc. An iconic model is used to describe the characteristics of the system rather than explaining the system. This means that such models are used to represent system's characteristics that are not used in determining or predicting effects that take place due to certain changes in the actual system.

(ii) Analogue models: An analogue model does not resemble physically the system they represent, but retain a set of characteristics of the system. Such models are more general than iconic models and can also to be manipulated

2. Symbolic models: These models use algebraic symbols (letters, numbers) and functions to represent variables and their relationships for describing the properties of the system. Such relationships can also be represented in a physical form. Symbolic models are precise and abstract and can be analysed by using laws of mathematics.

Symbolic models are classified into two categories.

(i) Verbal Models: These models describe properties of a system in written or spoken language. Written sentences, books, etc., are examples of a verbal model.

(ii) **Mathematical Models:** These models use mathematical symbols, letters, numbers and mathematical operators (+, −, ÷, ×) to represent relationships among variables of the system to describe its properties or behaviour. The solution to such models is obtained by applying suitable mathematical technique.

B. CLASSIFICATION BASED ON FUNCTION (OR PURPOSE)

1. Descriptive models: These models are used to investigate the outcomes or consequences of various alternative courses of action (strategies, or actions). Since these models evaluate the consequence based on a given condition (or alternative) rather than on all other conditions, there is no guarantee that an alternative selected is optimal. These models are usually applied (i) in decision-making where optimization models are not applicable, and (ii) when objective is to define the problem or to assess its seriousness rather than to select the best alternative. These models are especially used for predicting the behaviour of a particular system under various conditions. Simulation is an example of a descriptive model for conducting experiments with the systems based on given alternatives.

2. Predictive models: These models represent a relationship between dependent and independent variables and hence measure ‘cause and effect’ due to changes in independent variables. These models do not have an objective function as a part of the model of evaluating decision alternatives based on outcomes or pay off values. Also, through such models decision-maker does not attempt to choose the best decision alternative, but can only have an idea about the possible alternatives available to him.

3. Normative (or Optimization) Models: These models provide the ‘best’ or ‘optimal’ solution to problems using an appropriate course of action (strategy) subject to certain limitations on the use of resources. For example, in mathematical programming, models are formulated for optimizing the given objective function, subject to restrictions on resources in the context of the problem under consideration and non-negativity of variables. These models are also called *prescriptive models* because they prescribe what the decision maker ought to do.

C. CLASSIFICATION BASED ON TIME REFERENCE

1. Static models: Static models represent a system at a particular point of time and do not take into account changes over time. For example, an inventory model can be developed and solved to determine an economic order quantity assuming that the demand and lead time would remain same throughout the planning period.

2. Dynamic models: Dynamic models take into account changes over time, i.e., time is considered as one of the variables while deriving an optimal solution. Thus, sequences of interrelated decisions over a period of time are made to select the optimal course of action in order to achieve the given objective. Dynamic programming is an example of a dynamic model.

D. CLASSIFICATION BASED ON DEGREE OF CERTAINTY

1. Deterministic models: If all the parameters, constants and functional relationships are assumed to be known with certainty when the decision is made, the model is said to be deterministic. Thus, the outcome associated with a particular course of action is known, i.e. for a specific set of input values, there is only one output value which is also the solution of the model. Linear programming models are example of deterministic models.

2. Probabilistic (Stochastic) Models: If at least one parameter or decision variable is random (probabilistic or stochastic) variable, then the model is said to be probabilistic. Since at least one decision variable is random, the independent variable, which is the function of dependent variable(s), will also be random. This means consequences (or payoff) due to certain changes in the independent variable(s) cannot be predicted with certainty. However, it is possible to predict a pattern of values of both the variables by their probability distribution. Insurance against risk of fire, accidents, sickness, etc., are examples where the pattern of events is studied in the form of a probability distribution.

E. CLASSIFICATION BASED ON METHOD OF SOLUTION OR QUANTIFICATION

1. Heuristic models: If certain sets of rules (may not be optimal) are applied in a consistent manner to facilitate solution to a problem, then the model is said to be Heuristic.

2. Analytical models: These models have a specific mathematical structure and thus can be solved by the known analytical or mathematical techniques. Any optimization model (which requires maximization or minimization of an objective function) is an analytical model.

3. Simulation models: These models have a mathematical structure but cannot be solved by the known mathematical techniques. A simulation model is essentially a computer-assisted experimentation on a mathematical structure of a problem in order to describe and evaluate its behaviour under certain assumptions over a period of time. Simulation models are more flexible than mathematical models and can, therefore, be used to represent a complex system that cannot be represented mathematically. These models do not provide general solution like those of mathematical models.

IX. APPLICATIONS OF OPERATIONS RESEARCH

Some of the industrial/government/business problems that can be analysed by the OR approach has been arranged by functional areas as follows:

1. Finance and Accounting

- Dividend policies, investment and portfolio
- Break-even analysis, capital budgeting, management, auditing, balance sheet and cash cost allocation and control, and financial flow analysis planning
- Claim and complaint procedure, and public
- Establishing costs for by-products and accounting developing standard costs

2. Marketing

- Selection of product-mix, marketing and export
- Sales effort allocation and assignment planning
- Launching a new product at the best possible time
- Advertising, media planning, selection and
- Predicting customer loyalty effective packing alternatives

3. Purchasing, Procurement and Exploration

- Optimal buying and reordering with or
- Replacement policies without price quantity discount
- Bidding policies
- Transportation planning
- Vendor analysis

4. Production Management

- **Facilities planning**
 - Location and size of warehouse or new plant,
 - Logistics, layout and engineering design distribution centres and retail outlets
 - Transportation, planning and scheduling
- **Manufacturing**
 - Aggregate production planning, assembly
 - Employment, training, layoffs and quality line, blending, purchasing and inventory control
 - Allocating R&D budgets most effectively
- **Maintenance and project scheduling**
 - Maintenance policies and preventive
 - Maintenance crew size and scheduling maintenance
 - Project scheduling and allocation of resources
- **Personnel Management**
 - Manpower planning, wage/salary
 - Negotiation in a bargaining situation administration
 - Skills and wages balancing
 - Designing organization structures more
 - Scheduling of training programmes to maximize effectively skill development and retention
- **Techniques and General Management**
 - Decision support systems and MIS
 - Making quality control more effective forecasting
 - Project management and strategic planning
- **Government**
 - Economic planning, natural resources, social
 - Urban and housing problems planning and energy
 - Military, police, pollution control, etc.

Techniques of operations research

1. Linear Programming

It helps to find the best result (like maximum profit or minimum cost) using limited resources. Used when there are conditions or limits (constraints) to follow.

2. Transportation Problem

This technique finds the cheapest way to transport goods from suppliers to customers. It ensures supply meets demand at the lowest possible cost.

3. Assignment Problem

It assigns tasks or people in the most efficient way to save time or cost. Useful for job matching, like assigning workers to machines.

4. Game Theory

Used when competitors are involved to find the best winning strategy. It helps make smart decisions in competitive situations.

5. Queuing Theory

It studies waiting lines to reduce customer waiting time.
Helps in managing service counters or staff in busy areas.

6. Inventory Control

It decides how much stock to keep and when to reorder.
Prevents overstocking or running out of goods.

7. Simulation

It creates a model of a real situation to test ideas without risks.
Used to see results before trying them in real life.

8. Decision Theory

It helps in choosing the best option when the future is uncertain.
Useful for making smart business choices.

Evolution of Operations Research**1. Origin during World War II**

- OR began during World War II to solve military problems.
- Scientists used it to improve strategies like radar use, troop movement, and supply chains.

2. Post-War Development (1940s–1950s)

- After the war, OR was used in industries and business.
- Companies used it for production, transportation, and inventory control.

3. Growth in Business Applications (1950s–1960s)

- OR spread to fields like finance, marketing, and operations.
- Techniques like Linear Programming and Queuing Theory were developed.

4. Computer Era Boost (1960s–1970s)

- With the invention of computers, OR models became faster and easier to solve.
- Complex problems could now be handled using software.

5. Expansion to Service Sectors (1980s onwards)

- OR started being used in hospitals, airlines, banks, and public services.
- Helped in planning, scheduling, and resource management.

6. Modern OR (2000s–Present)

- OR is now used with Artificial Intelligence, Data Analytics, and Machine Learning.
- Widely applied in smart cities, e-commerce, logistics, and healthcare.